

Gaurav Chintakunta

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EDUCATION

University of Illinois at Chicago | Master of Science in Computer Science | GPA - 3.86/4.0 Expected May 2026
Coursework: Responsible AI Engineering, Neural Networks, Quantum Computing, Distributed Computing

JNTU Hyderabad | Bachelor of Technology in Computer Science May 2024
Coursework: Machine Learning, Design and Analysis of Algorithms, Data Science, Big Data

TECHNICAL SKILLS

- Languages:** Python, Go, Scala, JavaScript, SQL, C
- Frameworks & Tools:** FastAPI, React, Kubernetes, Docker, CrewAI, Git, GitHub Actions, pytest, Linux, Bash
- ML & AI:** PyTorch, TensorFlow, LangChain, Pinecone, RAG, LLMs, Transformers, PEFT, DSPy, Scikit-learn
- Data & Cloud:** Pandas, NumPy, Spark, OpenCV, Tableau, AWS, Azure, Streamlit
- CS Fundamentals:** Data Structures & Algorithms, System Design, Object-Oriented Design, Distributed Systems

EXPERIENCE

- Graduate Researcher** Jan 2025 - Present
University of Illinois at Chicago, Computer Science Chicago, IL
 - Boosted spatial prediction accuracy by 30% by designing Pix2Path, a Transformer-based GAN pipeline processing 2TB+ of spatialomics data on UIC's Lakeshore HPC cluster via optimized Docker workflows.
 - Reduced model epoch training time by 40% by engineering parallelized patch extraction and memory-efficient data ingestion pipelines using Python and Spark.
 - Cut experiment setup overhead and improved reproducibility across the research team by building automated TensorBoard tracking and custom checkpointing across multi-GPU training runs.
 - Accelerated architecture iteration across 5+ GAN variants by designing modular, reusable OOP pipeline components in Python, eliminating codebase restructuring between experiments and shortening the research feedback loop.
- Machine Learning Intern** May 2023 - Sep 2023
AIML Analytics Solutions Hyderabad, Telangana
 - Forecasted key business performance metrics with 15% accuracy improvement by building and validating Scikit-learn models against real-time SQL data streams.
 - Built and maintained RESTful data validation services integrated with SQL backends, eliminating data inconsistencies across model training and evaluation pipelines and reducing pipeline failures.
 - Integrated ML model outputs into Tableau business dashboards used by stakeholders across 3 teams, cutting the decision-making cycle by 20%.

PROJECTS

- Autonomous Agentic SRE Pipeline | Go, Python, Kubernetes, CrewAI, Gemini** [GitHub]
 - Engineered an event-driven microservices system deploying a Go watchdog agent inside Kubernetes to continuously poll Prometheus telemetry and evaluate anomaly thresholds in real time.
 - Architected a cross-network webhook pipeline routing alerts to an external Python Flask orchestrator, decoupling AI diagnostic logic from cluster infrastructure and minimizing in-cluster resource overhead.
 - Integrated CrewAI and Google Gemini to autonomously diagnose container memory spikes and generate precise kubectl rollout remediation commands, achieving full closed-loop Agentic AI automation with zero human intervention.
- AI Research Assistant | Python, LangChain, FastAPI, Pinecone, RAG** [GitHub]
 - Built a production-ready RAG API enabling researchers to query academic documents in natural language, sustaining sub-200ms latency under concurrent load with 95% retrieval consistency across CI/CD deployments.
 - Optimized Pinecone vector indexing and metadata filtering strategies to improve semantic retrieval accuracy across large document corpora; automated regression testing via GitHub Actions across iterative deployments.
- AI Documentation Bot | Python, LangChain, Pinecone, OpenAI** [GitHub]
 - Built a semantic search tool enabling developers to query Pandas, NumPy, and Scikit-learn documentation in plain English across 3 large technical corpora.
 - Integrated an MLP classifier to rank results by source reliability, improving answer quality by prioritizing official API references over community content.
- Movie Recommendation Engine | Python, Docker, FastAPI, SVD++** [GitHub]
 - Achieved 0.92 RMSE on a personalized movie recommendation system using SVD++ collaborative filtering, with an automated ML lifecycle covering training, validation, and offline model retraining.
 - Containerized the inference engine with Docker and exposed it via a FastAPI REST endpoint, sustaining sub-200ms latency under simulated concurrent load testing.